

Sai Prabhur Reddy Meka

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Data Scientist with **6 years** of experience in AI, specialising in Generative AI ,machine learning, deep learning, NLP, cloud deployments, and healthcare solutions. Expertise in developing scalable AI solutions across various domains, including generative AI, computer vision, healthcare, human resource (HR) automation, and autonomous vehicle systems.

SKILLS & EXPERTISE

- **Technical Skills:**

- **Programming Languages:** Python
- **Cloud:** AWS,AZURE
- **Frameworks & Libraries:** CrewAI, LangChain, LangGraph,Temporal,Django, FastAPI, Codeact and React,Long term memories, short term memories (Mem0 Framework),Daytona
- **Machine Learning & Deep Learning:** Machine Learning, Deep Learning, Computer Vision, CNN, LSTM, RNN, Encoder-Decoder, BERT, Generative AI, Large Language Models (LLMs), LangChain, LLM Agents, AI Agents, CrewAI.
- **Natural Language Processing:** Named Entity Recognition (NER), BERT, LLM Agents, Retrieval-Augmented Generation (RAG), OpenAI API,Claude,.
- **Computer Vision:** Image Processing, Object Detection, and Segmentation.
- **Tools & Platforms:** Docker, Git, GitLab, Subversion, Weights & Biases (WandB),EC2,ECR.

- **Languages Known:**

- English (Fluent), Telugu (Native), Malayalam (Fluent).

PROFESSIONAL EXPERIENCE

6 YEARS

GRID DYNAMICS

Data Scientist

1 Project : AI AgentFactory.

Hyderabad

APR 2025-Present

The Agent Factory project is a platform that allows businesses to easily create, deploy, and manage AI agents for automating complex tasks and workflows. It helps organisations build intelligent, collaborative agents that can perform data analysis, decision-making, and process automation without extensive manual setup.

- Designed and architected the Agent Factory platform to enable scalable creation, deployment.
- Integrated Generative AI, Machine Learning models, and workflow orchestration frameworks such as LangGraph and Temporal to build adaptive, intelligent, and automated agent pipelines
- Developed core system modules and reusable software components to standardise architecture, improve scalability, and significantly reduce time-to-deployment for new agents.
- Implemented advanced memory management systems using the Mem0 platform integrated with Azure PostgreSQL to provide both short-term and long-term memory for AI agents, enabling context retention and enhanced conversational intelligence.
- Built and deployed MCP (Model Context Protocol) servers to handle CRUD operations for memory, search tools, and image generation modules, ensuring smooth agent interaction with dynamic resources and external services.
- Implemented a CodeAct-based code generation and execution workflow using Daytona, enabling agents to generate, execute, and iteratively refine long-form code solutions dynamically.
- Conducted in-depth research and proof-of-concept (POC) experiments to enhance the platform's decision-making capabilities, intelligence, and overall adaptability.
- Collaborated closely with key stakeholders to gather technical and business requirements, ensuring the platform's design and functionality aligned with organisational goals and client needs.

Skills : Python,LangGraph, Temporal, Mem0, Azure PostgreSQL, MCP, Docker, FastAPI, Daytona, CodeAct, Generative AI, RAG, LLMs, Software Architecture, Workflow Orchestration

1 Project : ADAS & ECU Intelligent Search Assistant

Developed Built a RAG-based assistant that lets ADAS/ECU engineers instantly search technical knowledge like DTCs, CAN signals, and calibration details using natural language.

- Designed and deployed a RAG-powered Technical Knowledge Assistant for Honda's ADAS & ECU teams, enabling natural-language search across DTC definitions, CAN signals, calibration parameters, testing procedures, and ECU interface specifications.
- Built a scalable retrieval pipeline that ingests and indexes large volumes of engineering documents, making technical information easily searchable.
- Implemented natural-language query support so engineers can quickly find signal meanings, error codes, and subsystem interactions without manually searching through documents.
- Integrated retrieval to ensure high accuracy and relevant answers for complex technical questions.
- Reduced engineering lookup and debugging time from hours to seconds, speeding up ADAS/ECU development cycles and reducing dependency on domain experts.

Skills :Python, RAG, LLMs, Azure, Docker, Vector Databases, Embeddings, Prompt Engineering..

2 Project : AI Industrialisation.

Enables autonomous vehicles to perceive surroundings by fusing camera, LiDAR, and radar data for object detection and safe navigation.

- Worked on optimising and quantising the models using in build Software Development Kit of Qualcomm and worked on the post processing module for the perception stack.
- Built C++ bridge modules to support integration of custom ROS message formats with hardware interfaces, ensuring stability and compatibility across the perception stack.
- Collaborated with cross-functional teams to validate perception outputs and streamline deployment workflows under the AI Industrialisation program.

Skills : Python ,Computer vision, C++, Gitlab.

SYNAPSICA HEALTHCARE **Software Engineer AI**

Bangalore
SEP 2022-Aug 2023

1.Project : Spindle

To build an AI system which will detect the abnormalities in the spine and reduce the reporting time for radiologists and generate reports based on the spine diagnosis.

- Worked on binary classification problem to detect vertebral disc using semantic segmentation and achieved IOU of 0.87.
- Worked on the object detection task for detecting facet joints in spine MRI using Yolo and achieved a mAP of 0.89 .
- Enhancing the prediction data pipeline for the spindle project and building a new pipeline for a couple of pathologies and doing some geometrical enhancement for a couple of pathologies to get the predictions accurate.
- Worked on preparing MRI images for ground truth for the data team by aligning different sets of data to annotate in CVAT for providing json for AI models.
- Actively participated in the AWS deployment team, handling backend tasks using AWS services and SSH access.
- Managed EC2 instances by performing routine maintenance tasks, such as clearing cache and resolving issues when the backend code became unresponsive or hung.
- Leveraged AWS terminal (SSH) for troubleshooting, performing code rollbacks, and resolving runtime issues to ensure uninterrupted server operations.
- Wrote scripts using the Paramiko library to continuously monitor the server and ensure uptime, checking server status at regular intervals and automatically reinitiating server processes if any issues were detected, ensuring high availability and reliability of the backend systems.

Skills : Python,Deep learning,Computer vision, Open cv, AWS, Django, Gitlab.

2. Project : Spindle (product) Dashboard

Developing a comprehensive dashboard for effectively debugging an AI application spindle, allowing real-time tracking of the execution Row, comparison of outputs, monitoring of CPU utilisation, and identification of mis predictions in vertebra annotations made by the AI model.

- Accelerated a dashboard in a team of 3 using the tool python DASH which helps in debugging the main spindle AI backend server.
- Implemented a feature for comparing huge volumes of AI generated json data and Radiologist generated json data feature which eased the work of the AI scientists by implementing a unique feature of json difference highlighter.
- Implemented cpu and memory utilisation charts which dynamically updates while running the spindle AI backend which helps in knowing the utilisation of each backend module.

Skills : Python, Dash, Open cv, Dash, Flask, Gitlab.

HUSYS-A PEOPLE 2.0 COMPANY

Associate Technology

Hyderabad

MAR 2020-Sep 2022

1. Project : Automated Resume Parsing System

Processed and fine-tuned the model on a curated dataset of 4,000+ annotated resumes, across key entities such as skills, companies worked at, colleges attended, phone numbers, and email addresses..

- Designed a robust data pipeline to preprocess diverse resume formats (PDFs, Word documents) and handle noisy, unstructured data efficiently.
- Reduced manual resume parsing time by over 70%, enhancing productivity and accuracy for HR operations.
- Utilised Hugging Face Transformers, Python, and PyTorch to build and fine-tune the model, ensuring scalability for larger datasets and future enhancements..

Skills : Python, Transformers, Subversion. Project : Data Reports

1. Project : Data Reports

Data Reports is an application that fetches EPF, TDS, and Employee Salary Reports, Modules Status Reports, and generates reports by utilising XML-RPC calls from the HR application Aplusys. It facilitates the seamless uploading of reports to respective government portals and enables comprehensive business analysis.

- Working on generating EPF, TDS, Bank Reports, module status reports by using XML-RPC calls and writing scripts for fetching records from nearly 300 instances and merging it with an app build on django.
- Interacting directly with clients regarding feedback on reports generated, enhancement/modifying the existing code based on the client requirements.

Skills : Python ,Django. HTML, Odoo.

EDUCATION

SRI CHUNDI RANGANAYAKULU ENGG COLLEGE, JNTUK

Guntur, AP

- B Tech in Electrical and Electronics Engineering.

(2014-2018)